IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

- Executive Summary
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- Results
- Conclusion
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Executive Summary

Collected data from public SpaceX API and SpaceX Wikipedia page. Created labels column ‘class’ which classifies successful landings. Explored data using SQL, visualization, folium maps, and dashboards. Gathered relevant columns to be used as features. Changed all categorical variables to binary using one hot encoding. Standardized data and used GridSearchCV to find best parameters for machine learning models. Visualize accuracy score of all models.

Four machine learning models were produced: Logistic Regression, Support Vector Machine, Decision Tree Classifier, and K Nearest Neighbors. All produced similar results with accuracy rate of about 83.33%. All models over predicted successful landings. More data is needed for better model determination and accuracy.

Introduction

- Project Background :
  - Commercial Space Age is Here.
  - Space X has best pricing ($62 million vs. $165 million USD).
  - Largely due to ability to recover part of rocket (Stage 1).
  - Space Y wants to compete with Space X.

- Problems :
  - Space Y tasks us to train a machine learning model to predict successful Stage 1 recovery.

Section 1

Methodology

Methodology

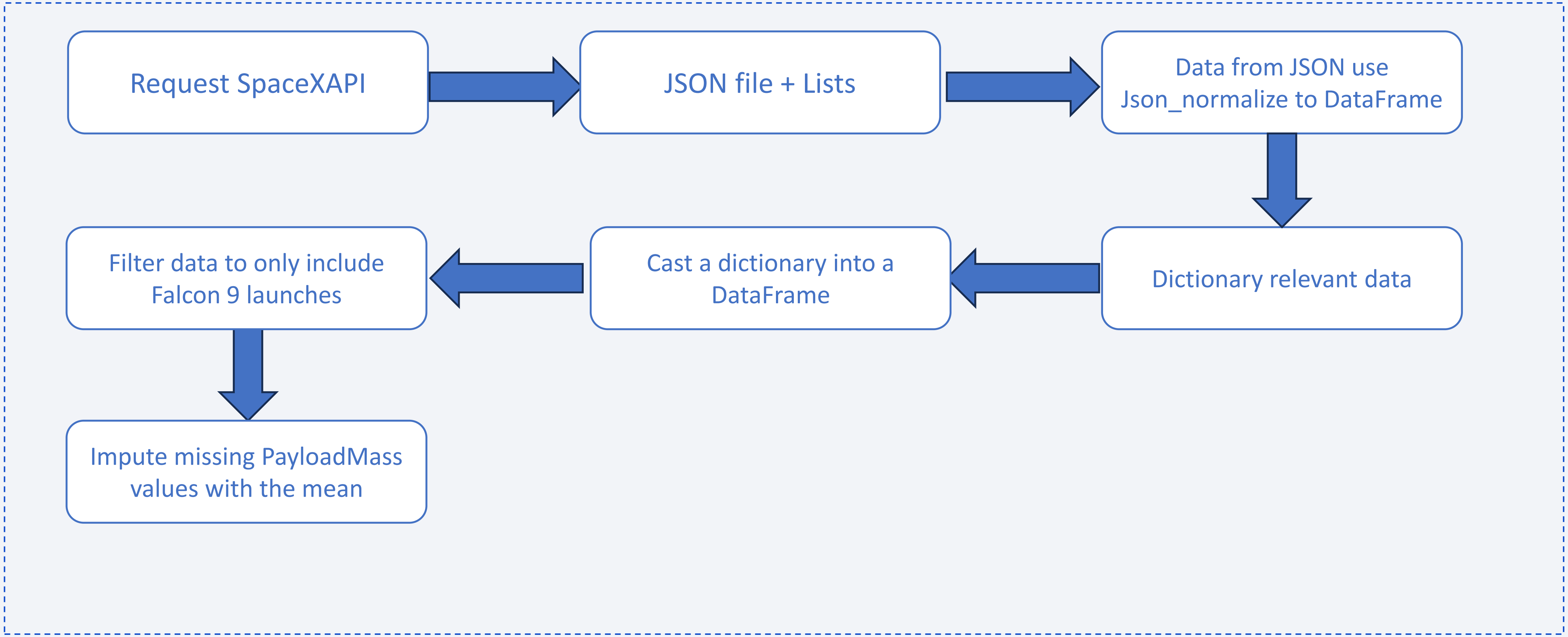
Executive Summary

- Data collection methodology:
  - Combined data from SpaceX public API and SpaceX Wikipedia page.
- Perform data wrangling
  - Classifying true landings as successful and unsuccessful otherwise.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Tuned models using GridSearchCV.

Data Collection

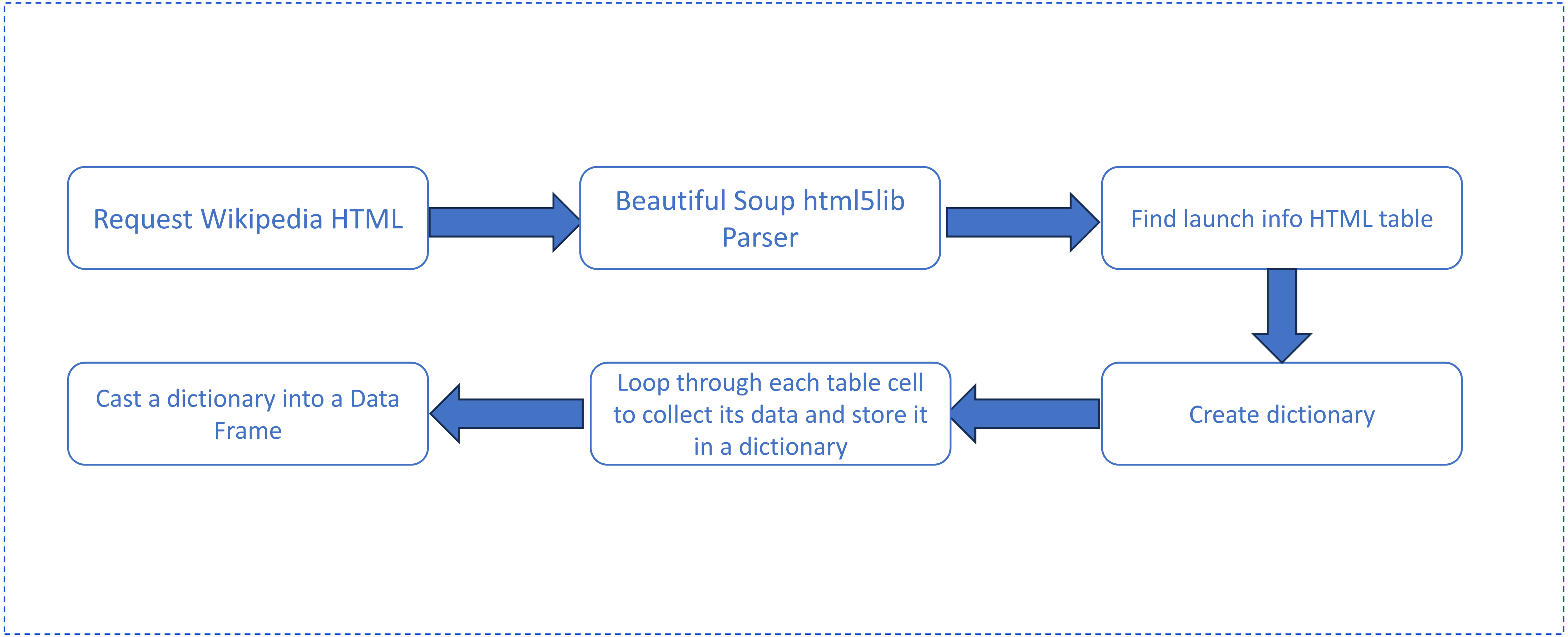
- Data Collection Overview:
  - Data collection process involved a combination of API requests from Space X public API and web scraping data from a table in Space X's Wikipedia entry.
  - The next slide will show the flowchart of data collection from API and the one after will show the flowchart of data collection from webscraping.
- Space X API Data Columns:
  - FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude,Latitude.
- Wikipedia Webscrape DataColumns:
  - Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time.

Data Collection – SpaceX API



- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/Data%20Collection%20Api%20.ipynb

Data Collection – Scraping



- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

- Create a training label for landing outcomes:
  - Success = 1, Failure = 0
- The Outcome column combines two fields: Mission Outcome and Landing Location.
- A new label column ‘class’ was created:
  - Value = 1 if Mission Outcome is True
  - Value = 0 otherwise
- Value Mapping:
  - True ASDS, True RTLS, True Ocean → 1
  - None None, False ASDS, None ASDS, False Ocean, False RTLS → 0

- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- EDA performed on Flight Number, Payload Mass, Launch Site, Orbit, Class, and Year.
  - Plots used:
  - Flight Number ⟷ Payload Mass
  - Flight Number ⟷ Launch Site
  - Payload Mass ⟷ Launc h Site
  - Orbit ⟷ Success Rate
  - Flight Number ⟷ Orbit
  - Payload ⟷ Orbit
  - Yearly Success Trend
- Charts: scatter, line, bar — to assess relationships for downstream ML features.

- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/edadataviz.ipynb

EDA with SQL

- Loaded the dataset into IBM Db2.
- Ran SQL via Python integration.
- Queries explored the dataset: launch-site names, mission outcomes, customer payload sizes, booster versions, and landing outcomes.

- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- Folium maps mark launch sites, successful/unsuccessful landings, and proximity to key features (railway, highway, coast, city). This helps explain site placement and shows landing outcomes relative to location.

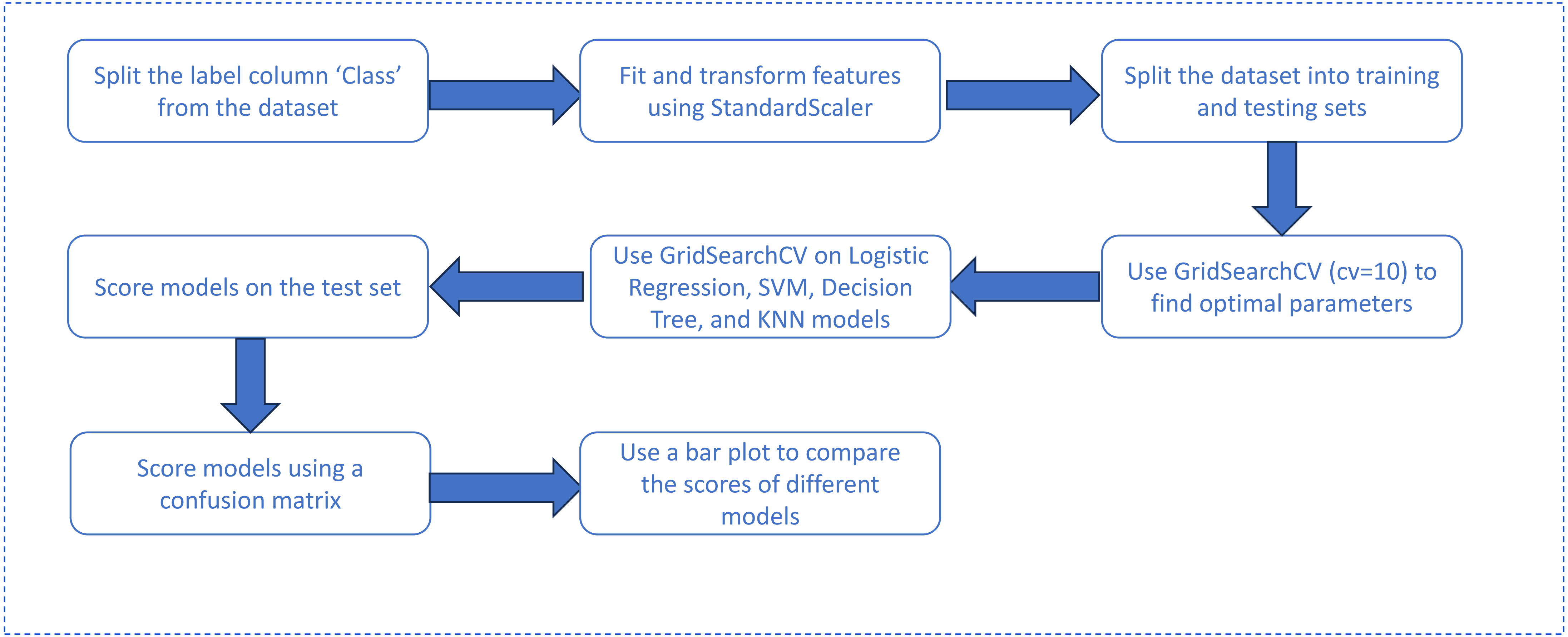
  - GitHub URL:
    - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- The dashboard includes a pie chart and a scatter plot.The pie chart can show either the distribution of successful landings across all launch sites or the success rate for an individual site.The scatter plot takes two inputs: site scope (all sites or a specific site) and payload mass via a 0–10,000 kg slider.The pie chart visualizes launch-site success rates, while the scatter plot reveals how success varies by site, payload mass, and booster-version category.

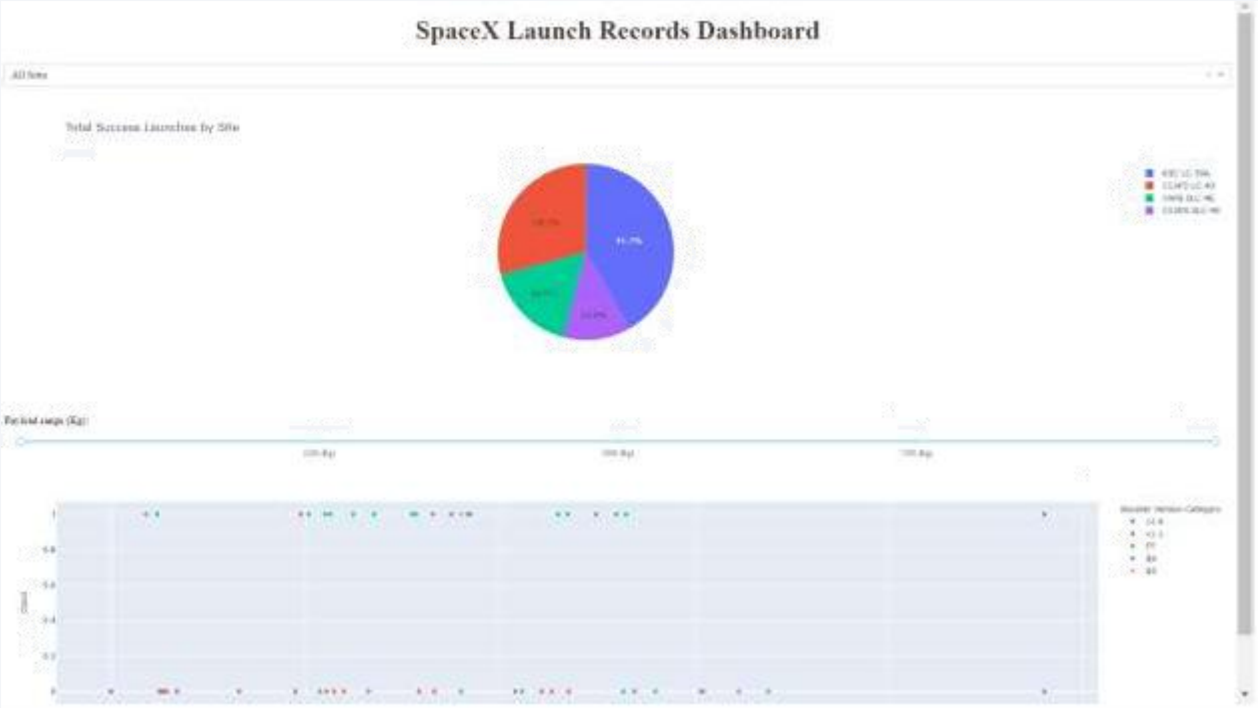
- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)



- GitHub URL:
  - https://github.com/ShangLin1606/IBM-Data-Science/blob/main/Machine%20Learning%20Prediction.ipynb

Results

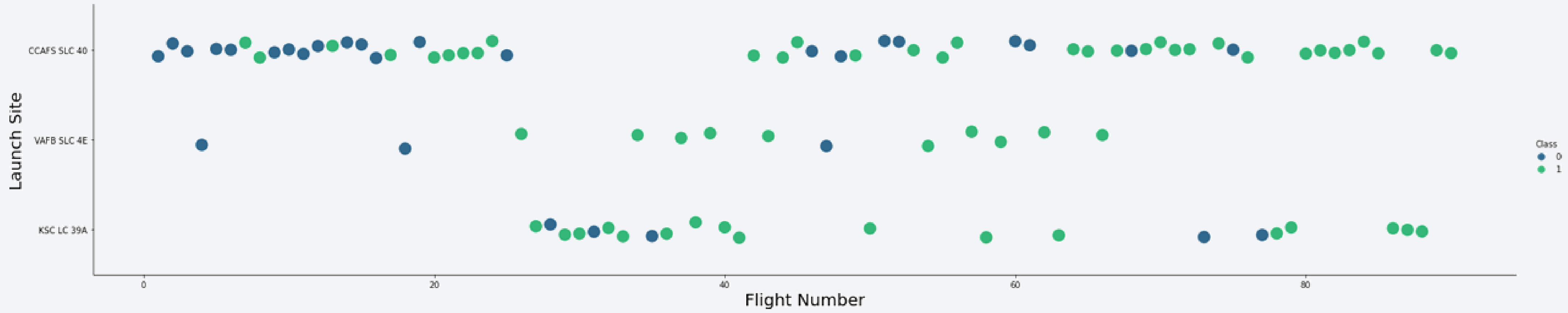


- This is a preview of the Plotly dashboard. The following slides will present the results of EDA with visualization, EDA with SQL, an interactive map built with Folium, and finally the model results, which achieved about 83% accuracy.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



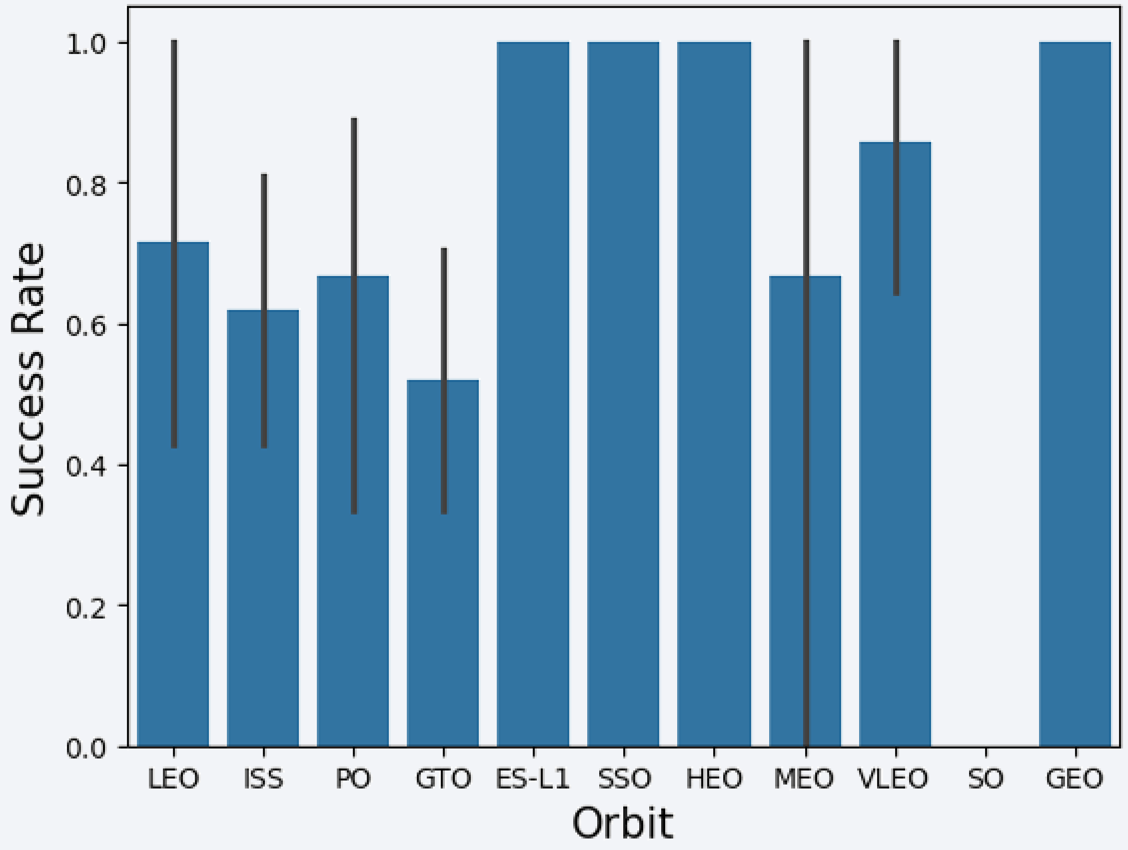
- Early flights (1–20): CCAFS SLC-40 was used most, with only two launches from VAFB SLC-4E; many early missions were unsuccessful, likely due to their experimental nature.
- Overall volume: CCAFS SLC-40 recorded the most launches. VAFB SLC-4E had the fewest (13 total, 3 failures).
- Trend: After the first 20 flights, success rates improved markedly across all sites.
- Late phase: Every launch after flight #80 was successful, occurring at CCAFS SLC-40 and KSC LC-39A.

Payload vs. Launch Site

- >80% of launches carried < 8,000 kg payloads — this is also where most failures occurred.
- There were 17 launches in the 8,000–16,000 kg range; only two were unsuccessful (VAFB SLC-4E and KSC LC-39A).
- CCAFS SLC-40 shows inconsistent outcomes as payload increases, whereas at KSC LC-39A nearly all launches below ~5,000 kg were successful.
- A notable cluster of failures appears between 5,000–6,000 kg, indicating a potential issue in that band.

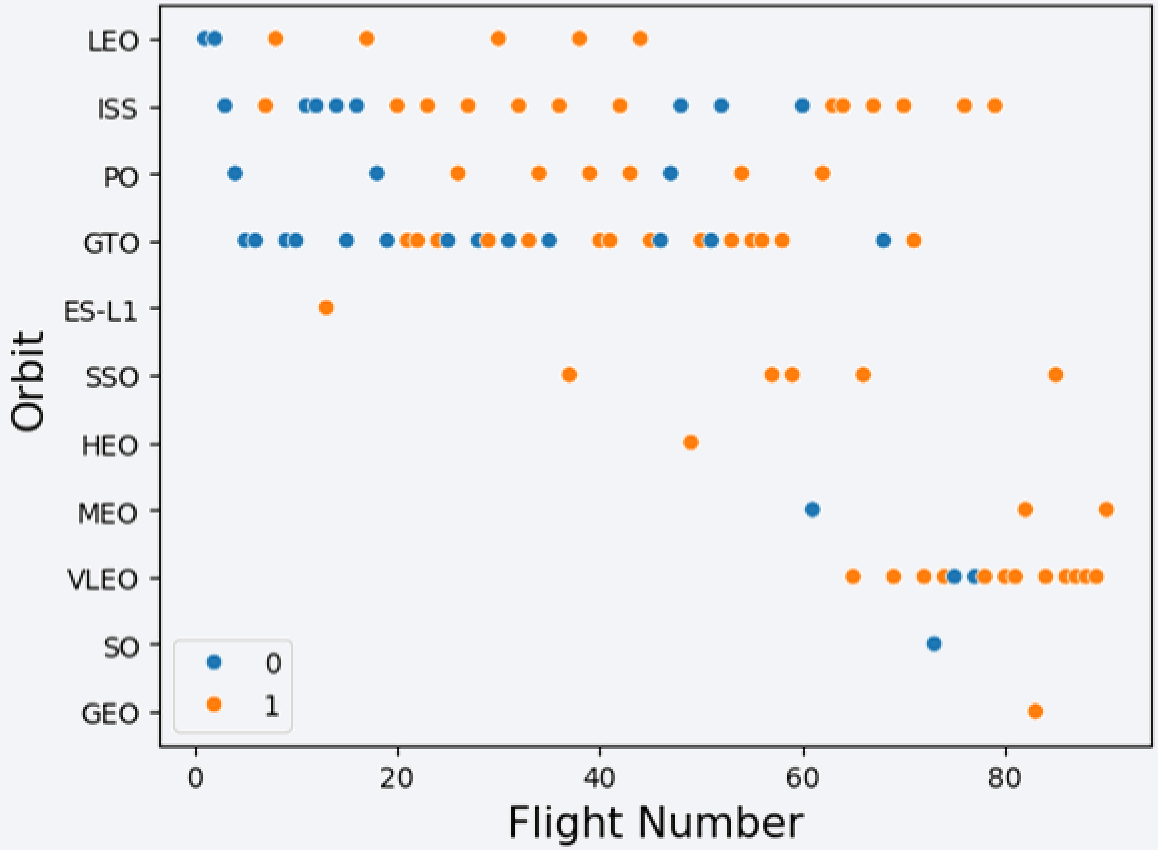
Success Rate vs. Orbit Type

- This chart shows the success rate by orbit for SpaceX launches.
- ES-L1, SSO, HEO, and GEO each show a 100% success rate.
- Note: The data are not normalized; you need the number of launches per orbit to judge statistical significance.



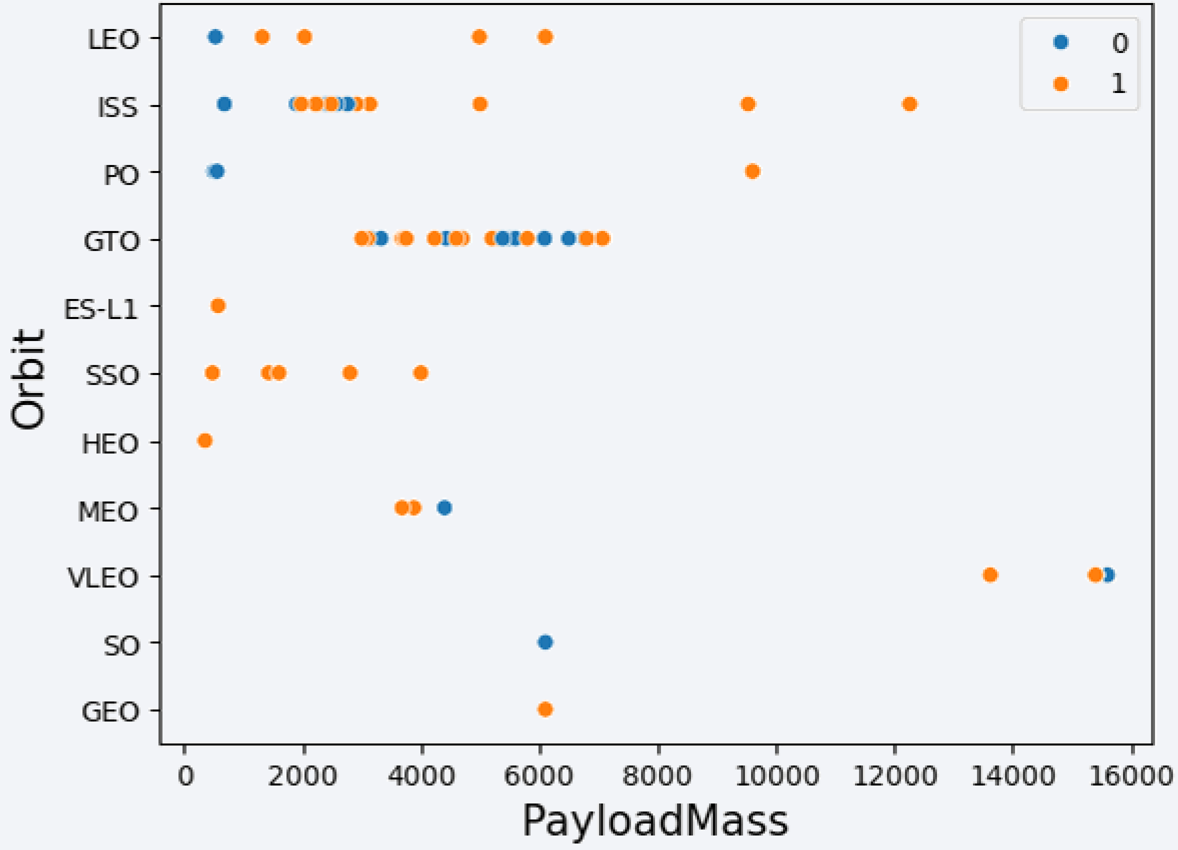
Flight Number vs. Orbit Type

- This chart shows the counts of successful and unsuccessful flights by orbit.
- The data are not normalized; about 70% of flights occurred in ISS, GTO, and VLEO orbits.
- Most failures happened in the first 20 launches.
- After 60 flights, success becomes nearly consistent across orbits.
- A marked increase in successes is visible after flight 20.



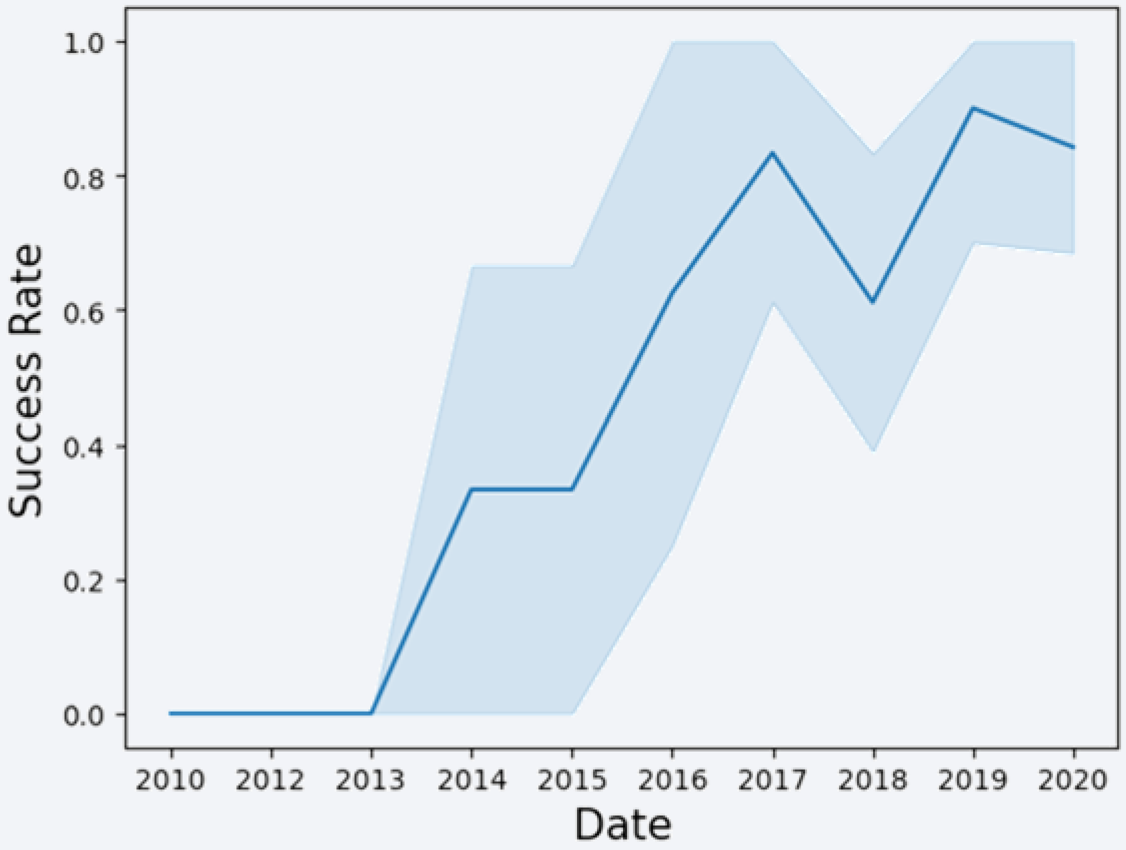
Payload vs. Orbit Type

- Most launches carry payloads under 8,000 kg.
- All SSO flights were successful, though most missions took place in ISS and GTO orbits.
- Failures cluster around ~500 kg and ~6,000 kg payloads.
- Payloads ≥ 8,000 kg show a higher likelihood of success.



Launch Success Yearly Trend

- The chart shows annual success rates from 2010 to 2020, with shaded bands indicating variability.
- 2010–2013: success rate remained at 0% (no successful launches).
- After 2013: the rate generally increases with new technologies, with a notable dip in 2018.



All Launch Site Names

- There are only 4 launch sites used

```
%sql select "Launch_Site" from SPACEXTBL group by "Launch_Site"
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Launch Site Names Begin with 'CCA'

```
%sql Select * from SPACEXTBL where "Launch_Site" Like 'CCA%' limit 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- This query finds 5 records where launch sites begin with `CCA`.

Total Payload Mass

```
%sql select SUM(PAYLOAD_MASS__KG_) from SPACEXTBL where "Customer"='NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
45596

- This query calculates the total payload carried by boosters from NASA

Average Payload Mass by F9 v1.1

```
%sql select AVG(PAYLOAD_MASS__KG_) from SPACEXTBL where "Booster_Version" like 'F9 v1.1%'
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

- This query calculates the average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

```
%sql select MIN(Date) from SPACEXTBL where "Landing_Outcome"='Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

- This query finds the date of the first successful landing outcome on ground pad.
- Since flights started in 2010, this means that it took almost six years until the first rocket succeeded.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql select "Booster_Version" from SPACEXTBL
    where "Landing_Outcome"='Success (drone ship)' AND PAYLOAD_MASS__KG_ between 4000 and 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Names of boosters that successfully landed on a drone ship (ASDS) with payloads between 4,000 and 6,000 kg.
- Only four boosters meet this criterion, indicating that successful landings in this payload band are relatively rare compared with other ranges.

Total Number of Successful and Failure Mission Outcomes

```
%sql select "Mission_Outcome",count("Mission_Outcome") from SPACEXTBL group by "Mission_Outcome"
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	count("Mission_Outcome")
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%sql select "Booster_Version" from SPACEXTBL where PAYLOAD_MASS__KG_=(select MAX(PAYLOAD_MASS__KG_) from SPACEXTBL)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- This is the list of the names of the boosters which carried the maximum payload mass. A subquery has been used in this SQL example.

2015 Launch Records

```
%%sql select substr(Date,6,2) as month, substr(Date,0,5) as year, "Booster_Version", "Launch_Site", "Landing_Outcome"
    from SPACEXTBL where "Landing_Outcome" like 'Fail%' AND substr(Date,0,5)='2015'
```

```
 * sqlite:///my_data1.db
Done.
```

month	year	Booster_Version	Launch_Site	Landing_Outcome
01	2015	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql select Date, count("Landing_Outcome"), "Landing_Outcome" from SPACEXTBL where Date between '2010-06-04' and '2017-03-20'
    group by "Landing_Outcome" order by Date DESC
```

* sqlite:///my_data1.db
Done.

Date	count("Landing_Outcome")	Landing_Outcome
2016-04-08	5	Success (drone ship)
2015-12-22	3	Success (ground pad)
2015-06-28	1	Precluded (drone ship)
2015-01-10	5	Failure (drone ship)
2014-04-18	3	Controlled (ocean)
2013-09-29	2	Uncontrolled (ocean)
2012-05-22	10	No attempt
2010-06-04	2	Failure (parachute)

Section 3

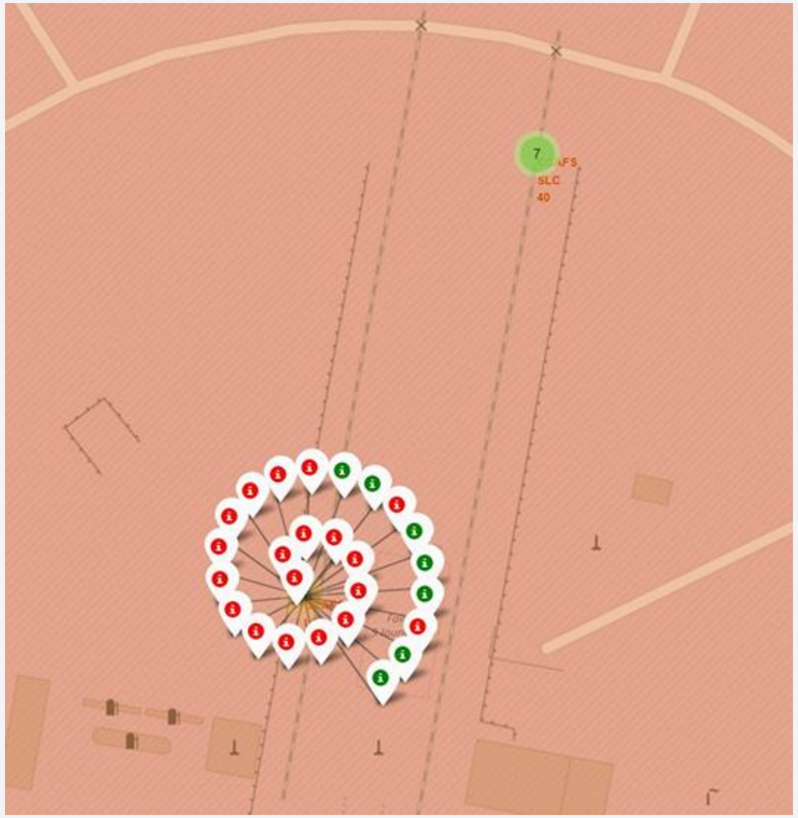
Launch Sites Proximities Analysis

Launch Sites Markers

- As shown, there are three launch sites on the East Coast of the United States and one on the West Coast.

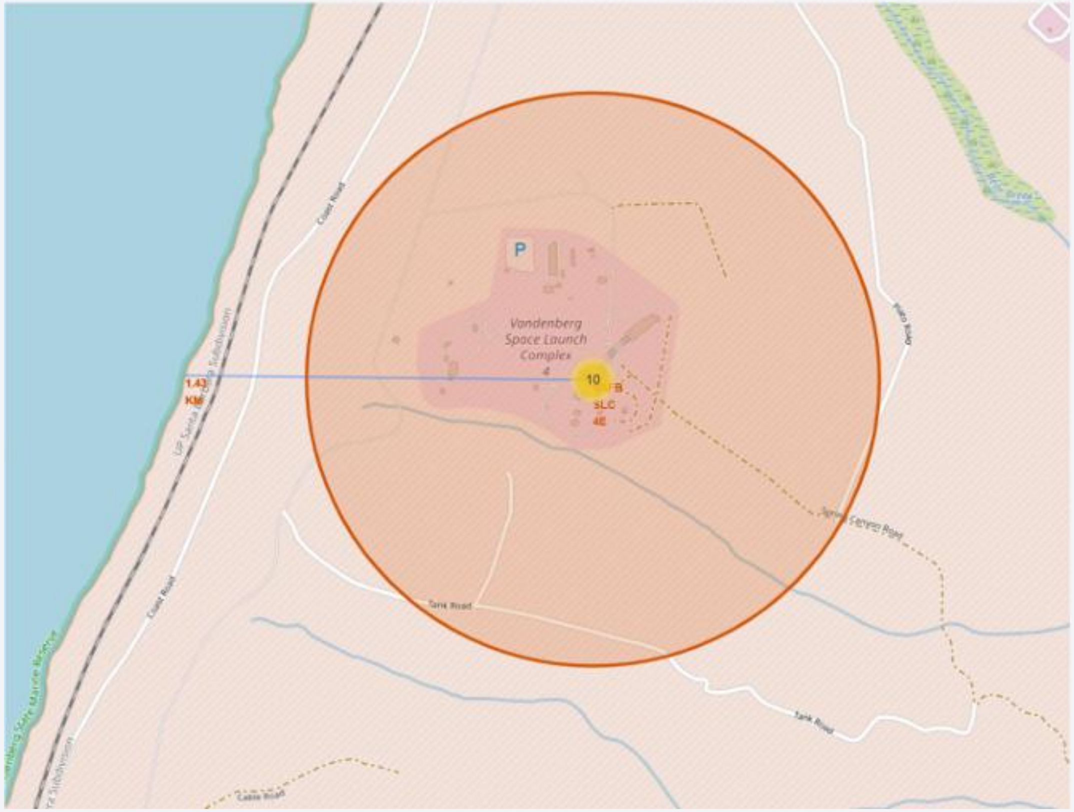
Success/Failed Launches For Each Site

- Color-coded launch outcomes are shown as marker clusters on the map.
- From the colors, we can quickly identify each site’s success rate.
- The map was built with the Folium library.



Distances Between a Launch Site To Its Proximities

- Distances to map features (highways, railways, airports, etc.) are computed by a Python function that uses mouse-pointer coordinates.
- In this example, the shortest distance from the launch-site center to the coastline is 1.43 km.

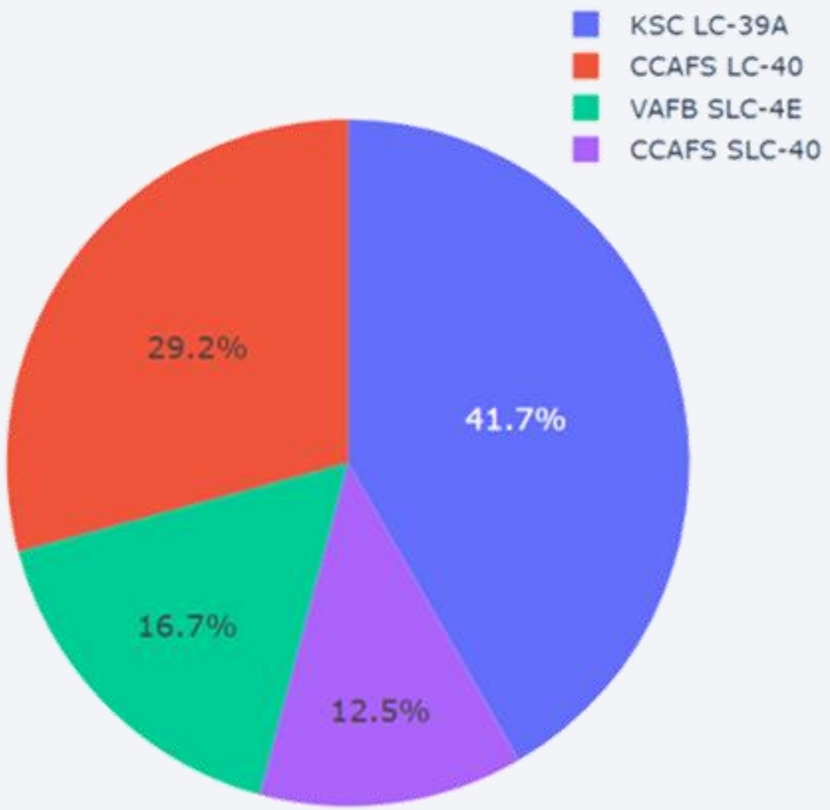


Section 4

Build a Dashboard with Plotly Dash

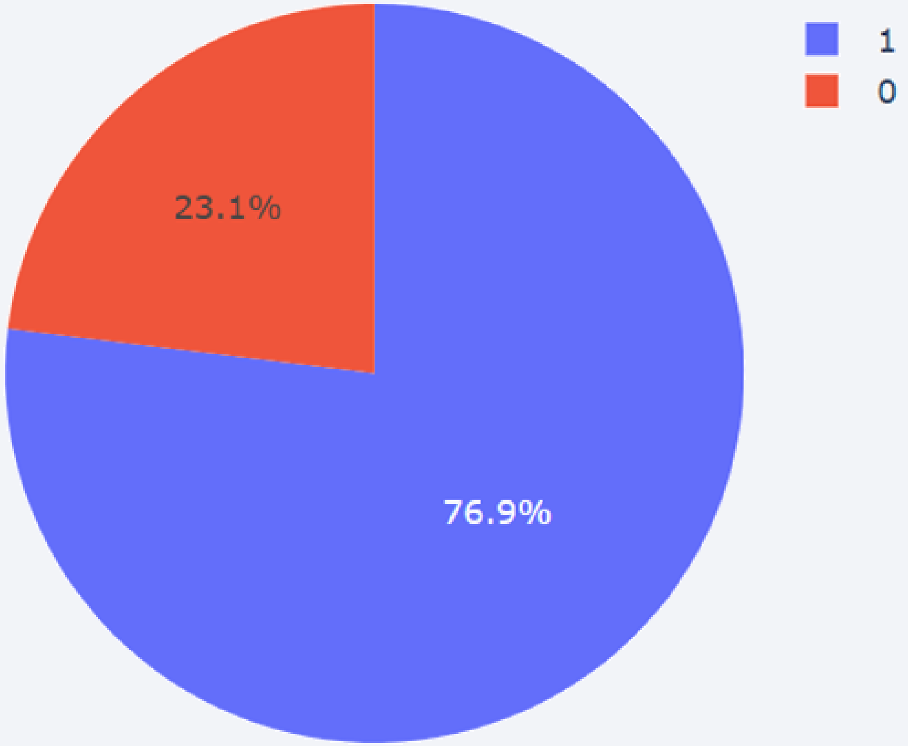
Total Successful Launches by Site

- Distribution of successful landings by site.
- CCAFS LC-40 → CCAFS SLC-40 (rename): CCAFS and KSC have the same count; most landings were pre-rename.
- VAFB has the smallest share, likely due to sample size and West Coast launch complexity.

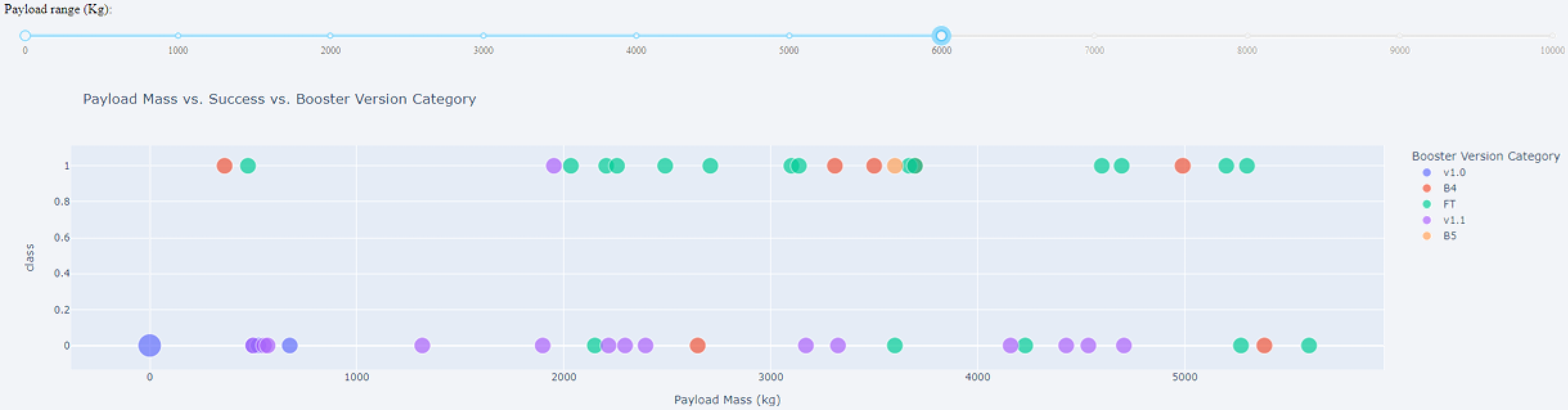


Success Rate For Site “KSC LC-39A”

- KSC LC-39A has the highest success rate, with 10 successful and 3 failed landings.



<Dashboard Screenshot 3>

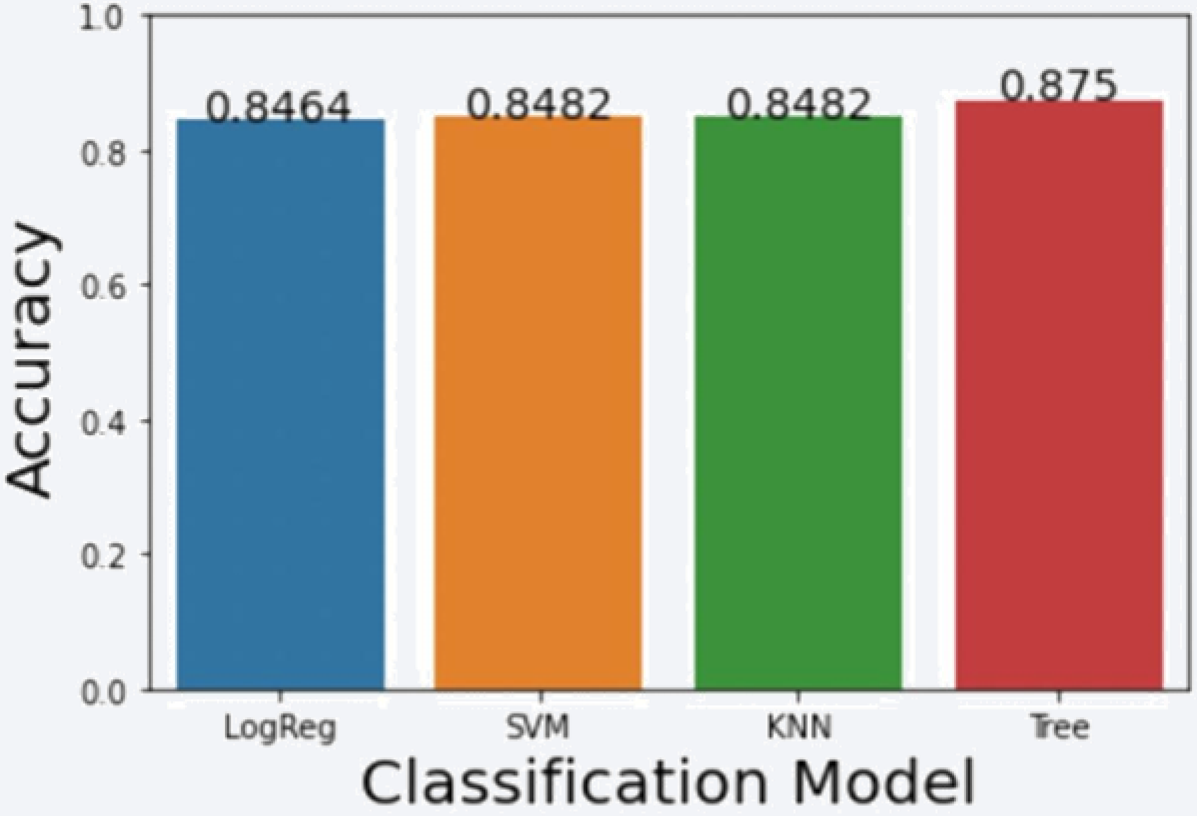


- Payload selector set to 0–10,000 kg (max payload = 15,600 kg).
- Class: 1 = success; 0 = failure.
- Scatter: color = booster version, size = launch count.
- In 0–6,000 kg, there are two failures at 0 kg.

Section 5

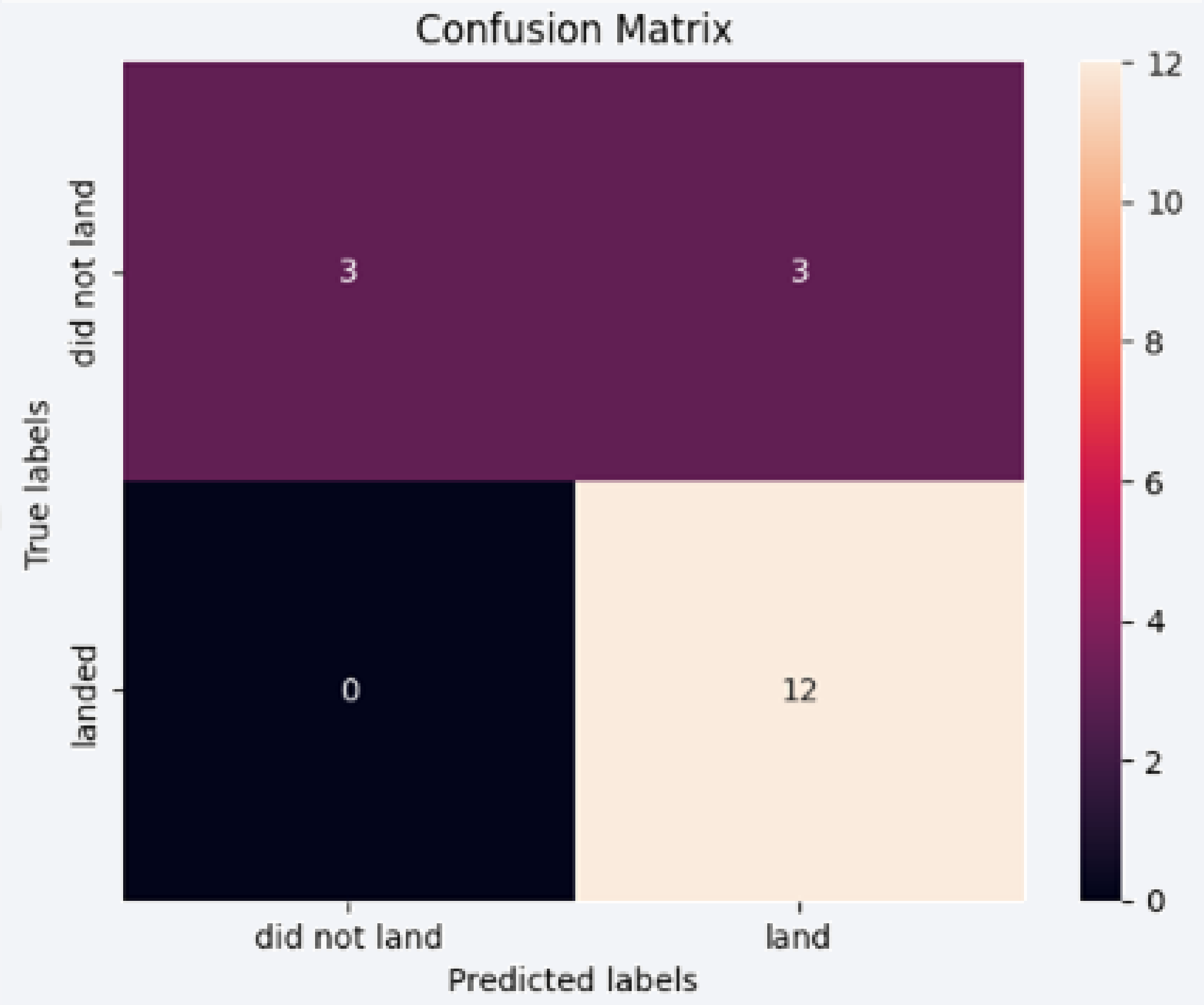
Predictive Analysis (Classification)

Classification Accuracy



- Average cross-validation scores (training set only) for all classifiers, shown as a bar chart.
- Model-building steps were described earlier.
- As shown, the Decision Tree achieves the highest average accuracy.

Confusion Matrix



- Precision : 1.0
- Recall: 0.5
- Accuracy: 83.33%

Conclusions

- During predictive analysis, multiple models converged to similar scores, largely due to class imbalance and data inconsistencies. Approximately 80% of records have a payload mass below 8,000 kg, which skews learning toward the majority class. In addition, outcomes labeled “no attempt” should be excluded from training and evaluation because they are not true landings.
- To improve accuracy, incorporate exogenous features such as weather conditions and vehicle/mission-level technical factors. With these enhancements, our analysis indicates which launch sites—and which site characteristics—are most associated with successful landings. We can use these site–property profiles to predict landing success more reliably, enabling better scheduling and resource allocation, and ultimately saving both time and cost.

Thank you!